Phase 7 TSP Operator Discovery Results (2026-05-20)

This note records the official 20-offset phase-7 modular operator-discovery campaign.

The official artifact archive for this phase is under runs/tsp_phase7_suite/operator_discovery/aggregate_20260520_085349_t.

Headline

Phase 7 produced real operator diversity and nontrivial rediscovery structure, but no operator survived the full validation pipeline.

- The campaign is technically valid: 20 paired runs, 800 modular epochs, zero modular generation fallbacks, zero materialization fallbacks, and zero missing_build_operator errors.
- full_solver_evolution was the best overall transfer condition.
- No modular condition beat full_solver_evolution on held-out TSPLIB gap.
- No discovered operator survived transplant, ablation, family-holdout, and Pareto validation simultaneously.

The correct interpretation is therefore that phase 7 acted as a strong falsification filter, not a positive operator-discovery result.

Main Findings

- Best mean overall condition: phase7_full_solver_evolution, with transfer gap 0.117916 and TSPLIB gap 0.164668.
- Best mean modular transfer condition: phase7_modular_operator_random_replay, with transfer gap 0.118833.
- Best mean modular TSPLIB condition: phase7_modular_operator_random_replay, with TSPLIB gap 0.230474.
- phase7_modular_operator_evolution versus phase7_baseline_heuristic_only: transfer delta -0.001715, 95% CI [-0.003418, 0.000085]; TSPLIB delta -0.003035, 95% CI [-0.010147, 0.000716].
- phase7_modular_operator_evolution versus phase7_full_solver_evolution: transfer delta 0.001686, 95% CI [-0.018321, 0.022827]; TSPLIB delta 0.067355, 95% CI [0.040498, 0.094769].
- phase7_modular_operator_random_replay versus phase7_modular_operator_evolution: transfer delta -0.000768, 95% CI [-0.002519, 0.000889].
- phase7_modular_operator_diversity_residual_replay versus phase7_modular_operator_random_replay: transfer delta 0.003009, 95% CI [-0.002884, 0.010797].
- phase7_modular_operator_compression_pressure versus phase7_modular_operator_evolution: transfer delta -0.000737, 95% CI [-0.004573, 0.002479]; novelty delta -0.244329.
- phase7_modular_operator_pareto_selection versus phase7_modular_operator_evolution: transfer delta 0.006411, 95% CI [0.001983, 0.012533]; runtime-inflation delta -1.605126.

Validation Outcome

No candidate satisfied the conservative surviving_candidate rule.

That means no operator simultaneously showed:

- positive transplant evidence across the scaffold set
- a clear family-level gain
- acceptable runtime/Pareto behavior
- ablation evidence that the operator itself mattered

The modular track therefore did not produce a reusable operator claim.

Rediscovery Outcome

Rediscovery was real, but it did not turn into validated discovery.

- Distinct rediscovery signatures: 56
- Most common signature: perturbation:double_bridge:segment_reversal:3:3
- Most common surviving signature: None

Repeated families included perturbation operators, restart controllers, candidate pruners, scaffold selectors, and some candidate-ranker variants. That suggests algorithmic attractors in the search space, but not validated reusable operators.

Interpretation

Phase 7 produced a negative or boundary result that is still scientifically useful.

- The modular operator pipeline is functioning as intended after the operator-specific generation-path fix.
- The pipeline is capable of generating diverse operator proposals and recurring operator families.
- Under the current scaffold set and validation rules, none of those operators justified a discovery claim.

This means phase 7 currently supports a stronger methodological claim than a positive discovery claim:

> the validation pipeline is strict enough to reject brittle or purely host-scaffold-specific operator artifacts.

That is a meaningful outcome for the research program, even though the ambitious discovery win condition was not met in this campaign.